

Department of Computer Science & Engineering

Indian Institute of Technology Kharagpur

Student Handbook

Executive Post Graduate Certificate in Generative AI & Agentic AI

32 Weeks | Online | Cohort-Based | 96 Hours Live by IIT KGP Faculty

Disclaimer

This handbook is an informational guide prepared in good faith to help learners find their way through the programme. It is not a legally binding contract and does not create enforceable rights or obligations beyond those set forth in the governing documents provided at the time of onboarding.

Programme structure, assessment frameworks, grading policies, and operational procedures described here reflect the best available information at the time of publication and may be revised during the course of the programme by IIT Kharagpur's Academic Committee. Where illustrative examples are provided (e.g., score calculations), they are for explanatory purposes only and do not guarantee specific assessment structures, weightings, or outcomes. In all academic matters, the decisions of IIT Kharagpur's Academic Committee are final.

How to Use This Handbook

- This handbook is your single reference for the Executive Post Graduate Certificate in Generative AI & Agentic AI, offered by the Department of Computer Science and Engineering at the **Indian Institute of Technology Kharagpur**, India's first IIT, established in 1951 and declared an Institute of National Importance.
- It covers programme structure, assessments, policies, support channels, and your rights and responsibilities. It is a living document. The online version on the LMS is always the most current. IIT Kharagpur reserves the right to update programme policies, assessment structures, and operational procedures as needed. Material policy changes affecting learner obligations or eligibility will be communicated through the LMS and official channels.

Key Policies at a Glance

Policy	What You Need to Know
Passing criteria	50% overall & 50% in proctored tests
Proctored tests	2 tests: one after Module 2, one at the end of Module 4
Capstone	Mandatory group project (max 5 per group); must meet rubric standards
Late submissions	Default 30% penalty till hard deadline (7 days from due date), from 8th day onwards, 100% penalty will be enforced
Deferral	One-time post-start till Module 1 deadline; 10% fee + immediate next batch
Plagiarism	Zero marks on the assessment; possible programme-level consequences
Support	studentsupport@upgrad.com
Escalation	iitkgpescalations@upgrad.com

1. Programme Overview

1.1 What This Programme Builds

- This programme takes you from GenAI prototypes to production-grade systems. The focus is end-to-end: from business problem definition to architecture selection to optimisation for cost, speed, and reliability to running a system in production where real users depend on it.

- IIT Kharagpur's philosophy here is engineering-first: build things that work reliably, understand why they work, and know what to do when they don't.

1.2 What You Are Expected to Know Coming In

The programme assumes you can write Python functions, work with APIs, read technical documentation, and are comfortable with the basic maths and statistics behind common ML concepts. If you need to brush up on any of these, do so before Week 1.

1.3 Learning Outcomes

Graduates who engage fully with the programme are expected to be able to:

- Explain how LLMs work at a first-principles level and select the right foundation model for a given use case
- Design and build enterprise-grade RAG systems with hybrid search, re-ranking, and measurable retrieval quality
- Fine-tune open-source LLMs using PEFT methods (LoRA/QLoRA), evaluate gains against a baseline, and determine when fine-tuning is worth the effort versus prompting or RAG
- Architect multi-agent systems with planning, tool use, memory, and orchestration
- Deploy GenAI systems to production with model serving, monitoring, latency/cost optimisation, and safety guardrails
- Justify architecture decisions with data and demonstrate understanding of trade-offs across cost, accuracy, latency, and scalability

Individual outcomes depend on prior experience, effort, and engagement with programme content.

2. Programme Structure

2.1 The Five Modules

1. **Module 1: Foundation Models & Transformer Architecture(Weeks 1-6)**
   - Deep learning essentials, transformer architecture (attention, tokenisation, embeddings, positional encoding), and working with foundation models (GPT, Gemini, LLaMA, Mistral). How to select the right model for cost, capability, and constraints.
2. **Module 2: Advanced Prompting & RAG Systems (Weeks 7-12)**
   - Systematic prompt patterns (tool-calling, retrieval-aware prompts, safety prompts, failure handling), RAG fundamentals (chunking, retrieval architectures, vector databases), and advanced RAG (hybrid search, re-ranking, evaluation frameworks, debugging retrieval failures).
3. **Module 3: LLM Fine-Tuning & Alignment (Weeks 13-18)**
   - When to fine-tune versus prompt versus RAG: a decision framework. Parameter-efficient fine-tuning (LoRA, QLoRA) on focused datasets. Three full weeks on PEFT methods. Hands-on lab sprint covering dataset prep, training runs, evaluation, and iteration.
4. **Module 4: Multimodal & Agentic AI (Weeks 19-24)**
   - Vision-language models and image generation pipelines. Agentic AI systems: planning, tool use, memory, orchestration, and multi-agent workflows.
5. **Module 5: Production Deployment & Industry Capstone (Weeks 25-30)**
   - Production-grade RAG and agent orchestration at scale. Model serving (vLLM-style concepts, FastAPI, containerisation), monitoring, and performance baselining. Responsible AI: guardrails, privacy, hallucination handling, cost control, and governance documentation. Industry capstone project.

3. Live Sessions, Academic Support & Recordings

3.1 Live Sessions

Detail	What to Expect
Schedule	Weekends, ~3 hours per session with breaks. A session calendar will be shared before each module, and invites will be sent in advance
Format	100% live by the IIT Kharagpur faculty
Total hours	Approximately 96 hours across 32 weeks
Attendance	Strongly recommended; no formal attendance criteria for certification *
TA presence	At least 1+ Teaching Assistant joins each session

3.2 Academic Support

- Your first line of academic support is the Teaching Assistants assigned by IIT Kharagpur. Academic queries are first directed to TAs; if unresolved, they are escalated to the faculty member.
- Faculty may share pre-read or reference material on or before the session day. Any additional reading resources will be uploaded to the LMS.
- Dedicated doubt resolution sessions are scheduled periodically across the programme to give you structured time to clarify concepts, revisit difficult topics, and get direct answers from Teaching Assistants and faculty.

3.3 Recordings

- Recordings of live sessions are made available on the LMS within 72 hours, subject to technical feasibility. Recordings are available for the duration of programme access.

4. Assessment & Grading

4.1 How You Are Assessed

The programme assesses you through three tracks:

Continuous assessment:

- Quizzes, hands-on projects, coding assignments, and module-level system-building projects. Assessment formats vary by module and may include MCQs, coding exercises, and case studies, as determined by the module's faculty.

Proctored assessment

Two proctored tests

Proctored Test	Timing	Covers
Test 1	After Module 2	Foundations, Prompting, RAG
Test 2	End of Module 4	LLM Fine-Tuning & Alignment, Multimodal, Agentic AI

- Each proctored test will be held on a weekend, with a primary slot and a backup slot on the following weekend. Exact dates and times will be published on the LMS calendar closer to

the exam, and invites will be shared in advance.
- Assessment weightages, once communicated for a module at its start, will not be changed for that module's assessments.

Capstone

- After Module 4, learners will be assigned an industry capstone project, which will serve as the final assessment component of the program.

4.2 Passing Criteria

Requirement	Threshold
Minimum overall score	50% across all assessments combined
Minimum in proctored tests	50% in proctored examinations
Module-level minimum	None, unless stated in a specific module brief
Capstone	Mandatory submission meeting rubric standards

4.3 Grading Scale

Grade	Grade Point
Excellent (EX)	10
Very Good (A)	9
Good (B)	8
Fair (C)	7
Average (D)	6
Pass (P)	5
Fail (F)	0

- **Overall CGPA = Weighted average of all component scores, normalised as per IIT Kharagpur's framework.**

5. Submissions and Deadlines

5.1 What You Submit

- Depending on the module: code implementations (RAG pipelines, fine-tuned models, agent workflows, deployment servers), project reports explaining architecture decisions and performance results, lab outputs, and capstone deliverables, including working prototypes and evaluation summaries.

5.2 Deadlines and Late Submission Policy

- All submissions must be completed before the LMS deadline timestamp. All assessment timelines are fixed and communicated in advance.
- The following default late submission framework, as approved by IIT Kharagpur's Academic Committee, applies unless a specific module brief states otherwise:

Submission Timing	Consequence
On or before due date	Full marks awarded based on the evaluation
Up to 7 days from due date	30% penalty on the assessed score
Beyond 7 days (hard deadline)	Submissions will receive feedback, but will be scored as zero for grading purposes

- Deadline extensions are not available under normal circumstances. In cases of documented medical or personal emergency, learners may submit a request to the Academic Committee through the Program Co-ordinator. Approval is at the Academic Committee's sole discretion and is not guaranteed. If a module brief specifies a different penalty structure for a particular assessment, that module-specific policy takes precedence for that assessment only.

5.3 Deadline Extension Policy

Deadlines for all graded components are strictly enforced and are considered non-negotiable, except under exceptional circumstances such as serious health issues or critical work-related obligations. In such cases, learners may be granted an extension of up to one week from the original deadline. During this extension period, submissions will not incur the standard 30% late penalty. Under no circumstances will extensions be granted beyond the hard deadline, which is defined as one week after the original submission date.
The following guidelines apply:

- Each learner is eligible to avail a maximum of three extensions throughout the duration of the program.
- Extensions may be applied to individual assignments, group case studies, and graded MCQs.
- No extension requests will be entertained beyond the hard deadline (one week after the original deadline).
- The option to apply for an extension remains available only until the hard deadline.
- Learners are expected to exercise discretion in using extensions and must apply them directly through the Learning Platform.
- For group case studies, each member must individually apply for an extension. Members who do not apply for an extension will be subject to late submission penalties if the group submission is made after the original deadline.
- Learners may resubmit assignments until the hard deadline; however, penalties will be applied based on the final submission timestamp.
- Extensions are not permitted for proctored examinations. These assessments will be conducted strictly as per the scheduled date and time.

5.4 Result Timelines

- Assessment result publication timelines will be communicated on the LMS for each graded component.

6. Academic Integrity and Responsible Use of AI

6.1 Academic Integrity

- Academic honesty is essential to maintaining IIT Kharagpur standards. Plagiarism includes: copying code or content without attribution, submitting another learner's work, unauthorised collaboration, and unpermitted use of AI tools beyond what is explicitly allowed for a given assessment.
- Confirmed violations may lead to zero marks on the assessment, disciplinary action, and possible programme-level consequences.

6.2 Responsible Use of Generative AI

Given this programme's focus on Generative AI, the use of AI tools must be clearly defined. The following guidelines have been approved by IIT Kharagpur's Academic Committee:

- **Acceptable use** (unless a specific assessment states otherwise): using AI tools to explore concepts, debug code, or accelerate research during self-paced learning; referencing AI-generated outputs as a starting point, with your own reasoning, modification, and attribution.
- **Unacceptable use:** submitting AI-generated work as your own without disclosure or meaningful modification; using AI tools during proctored tests; generating code, reports, or project deliverables primarily through AI tools without demonstrating your own understanding.
- Where is the line? Using an AI code-completion tool to autocomplete boilerplate code while you design the system architecture is acceptable. Using an AI text-generation tool to generate your entire RAG pipeline implementation and submitting it with minor variable renames is not, regardless of disclosure.
- **Important:** Disclosure does not convert otherwise prohibited use into permitted use. If the core logic, architecture, or implementation of a submission is generated by AI tools rather than developed by the learner, it may be treated as a violation regardless of disclosure.
- **Disclosure requirement:** Where an assessment permits AI assistance, you must disclose the tools used and the nature of their contribution. A brief statement listing the tool, the purpose of use, and the extent of assistance is sufficient.

6.3 Sharing Solutions

- Learners are not allowed to request, share, or discuss solutions for any graded component on any platform. First offence results in zero marks on that component. The programme administration reserves the right to delete posts containing answers without prior notice.
- Learners who become aware of shared solutions are expected to report the matter to their Programme Coordinator and must not use or retain the shared content.

7. Professional Conduct

Learners are expected to maintain professional and respectful communication across all programme platforms, including Telegram, discussion forums, and any other channels.

The following may result in disciplinary action, up to and including removal from the programme:

- Harassment, threats, abusive language, or any form of disrespect toward peers, faculty, or programme staff
- Any form of bullying, including sexual harassment
- Violations of academic integrity policies (Section 6)
- Sending bulk or mass communications to the programme team outside designated escalation channels

All programme-related communications should go through the official support channels listed in Section 9.

8. Deferral and Withdrawal

8.1 Pre-Programme Deferral

Aspect	Policy
Eligibility	Request before the Cohort Commencement Date
Approval	Subject to availability and the Institute's approval
Deferral fee	None
Programme fee	Differential amount payable if the deferred cohort fee is higher
Cohort availability	Subject to availability. If no future cohort exists, deferral is not possible

8.2 Post-Start Deferral

Aspect	Policy
Limit	One-time only; must be to a cohort starting within 1 year of original start, subject to Institute approval
Deferral fee	10% of the Total Programme Fee (inclusive of taxes), separate from and in addition to any differential fee. This fee cannot be adjusted against the Total Programme Fee.
Payment deadline	Within 7 days of the request; request expires if unpaid
Progress	All grades carry forward to the deferred cohort
Access	Original cohort access is disabled upon approval; learning resumes from the last graded assignment
Request deadline	Before the deferral cut-off date communicated for your cohort on the LMS
Until approval	You remain part of the original cohort and must continue as per programme requirements until the deferral fee is paid, and the request is approved Deferral availability is subject to future cohorts and the Institute's approval. If no future cohort is available, deferral is not possible

8.3 Withdrawal

Post-cohort refunds are not allowed. For pre-cohort refunds, please contact studentsupport@upgrad.com; or apply via the portal. Refer to the refund policy mentioned in the Offer Letter and program page.

9. Support, Communication & Escalation

9.1 Communication Channels

- **Telegram** is the official notification channel. It is a notification-only service; participants cannot post messages. Stay active and check regularly. Failure to stay up to date is not a valid excuse for missing deadlines.
- **For any queries**, please write to studentsupport@upgrad.com and/or you can also contact your Program Co-ordinator via Telegram (ID shared in your welcome email).

9.2 Escalation Framework

Level	Contact	Handles

Primary	studentsupport@upgrad.com	All first-contact queries
Level 1	iitkgpescalations@upgrad.com	Unresolved after primary
Level 2	escalation.l2@upgrad.com	Formal complaints and grievances
Level 3	escalation.l3@upgrad.com	Feedback and suggestions

- Always start at Primary. The support team endeavours to acknowledge queries promptly. Complex queries involving academic review may require additional time; learners will be kept informed.
- Academic grievances that remain unresolved after exhausting the escalation framework above may be referred to IIT Kharagpur's Academic Committee for final adjudication. The Programme Coordinator will facilitate this referral.

10. Certification

10.1 Your Certificate

Learners who meet all eligibility criteria are awarded an **Executive Post Graduate Certificate in Generative AI & Agentic AI** issued by **IIT Kharagpur**, subject to final academic approval by the institute.

10.2 Eligibility

- 50% overall score AND 50% in proctored tests
- Capstone submitted and meeting rubric standards
- Adherence to academic integrity and code of conduct policies
- All programme dues cleared

10.3 Certificate with Distinction

- The top 10% of the cohort may be awarded a **Certificate with Distinction**, as indicated on the credential.

10.4 Certificate Details

- The name on the certificate is printed as per enrolment records and cannot be modified after issuance. Certificates are issued by IIT Kharagpur; hard copies are dispatched by upGrad.
- A transcript summarising your grades and assessment performance is issued alongside the certificate.

11. Executive Alumni Status

- Learners who successfully complete the programme are eligible for **Executive Alumni status** from IIT Kharagpur.

Note

This Executive Education credential confers Executive Alumni status. It is not equivalent to a degree from IIT Kharagpur and does not confer the same alumni standing as degree programmes.

- Alumni may receive access to the IIT Kharagpur Executive Alumni Network, invitations to

academic events, and networking opportunities. All alumni benefits are subject to the institute's policies and may be time-bound or conditional.

12. Graduation Ceremony

The programme ends with a Graduation Ceremony at the IIT Kharagpur campus, subject to scheduling and administrative considerations. The institute may modify the format if circumstances require.

Detail	What to Know
Attendance	Optional; does not affect certification
Travel and accommodation	At the learner's expense
If you cannot attend	Your certificate will be couriered to your registered address after completion of all formalities

13. Re-Examination and Re-Evaluation

13.1 Re-Examination

- Learners who do not meet the minimum passing criteria may be offered a re-examination at the end of the programme (not per-module), subject to Academic Committee approval.
- Re-examinations, if offered, will typically take the form of a consolidated assessment covering the areas where the learner fell short. The format (written test, viva, project resubmission, or a combination) and any applicable fees will be communicated to affected learners within two weeks of final grade declaration.
- As per the Academic Committee's grading policy, re-examination scores are capped: the maximum achievable grade through re-examination is Pass (P), regardless of performance.
- If a learner does not pass after re-examination, they will not be eligible for certification. The Academic Committee's decision is final.

13.2 Re-Evaluation

- You may request re-evaluation of graded components within the timeline communicated alongside the result declaration on the LMS. Specify the component and justification clearly. Incomplete, vague, or unjustified requests will not be considered.
- Re-evaluation may result in an increase, decrease, or no change in marks. Requests for submissions beyond final deadlines may not be entertained. The Academic Committee's decision is final and binding.

14. What If...? Quick Reference

Situation	What Happens
I miss a live session	Watch the recording on LMS; no attendance penalty
I miss a proctored test	Use the backup slot (held the following weekend). No further extensions
I submit an assignment late	Default 30% penalty till hard deadline (7 days from due date), from 8th day onwards, 100% penalty will be enforced. Documented emergencies may be considered by the Academic Committee (see Section 5.2)

I failed a proctored test, but passed overall	You pass. The 50% proctored minimum is across proctored tests, not per-test
I fail overall	Re-examination may be offered at programme end; typically a consolidated assessment, capped at Pass grade (see Section 13.1)
I want to defer	A one-time deferral is allowed with a 10% fee. The request must be raised before the cohort's deferral cut-off and is subject to IIT KGP approval.
I want to withdraw	Contact studentsupport@upgrad.com; refund terms per upGrad T&Cs
I have a non-academic query	Write to studentsupport@upgrad.com first, then reach out to the Program Co-ordinator
I have an unresolved complaint	Escalate to iitkgpescalations@upgrad.com
I suspect plagiarism in my group	Report to Program Co-ordinator; do not retain or use shared content
I want to know where I stand	Use the score calculation method in Section 4.4 with your current scores

Who to Contact

I need help with...	Contact
LMS, tech support, payments, fees	studentsupport@upgrad.com
Session schedule, recordings, Telegram	Programme Coordinator (via Telegram)
Assessment queries, grading	studentsupport@upgrad.com (routed to academic team)
Deferral or withdrawal	studentsupport@upgrad.com
Unresolved issue	iitkgpescalations@upgrad.com
Formal complaint	escalation.l2@upgrad.com
Feedback and suggestions	escalation.l3@upgrad.com

Appendix A: Glossary

Term	Meaning
EPGC	Executive Post Graduate Certificate
LMS	Learning Management System (the programme's online platform)
RAG	Retrieval-Augmented Generation
LLM	Large Language Model
PEFT	Parameter-Efficient Fine-Tuning
LoRA / QLoRA	Low-Rank Adaptation / Quantised Low-Rank Adaptation
CGPA	Cumulative Grade Point Average
Capstone	The end-to-end system project in Module 5
Cohort	Your batch of learners is starting the programme together

Programme Coordinator	Your primary point of contact for programme operations
Deferral cut-off date	The final date by which a deferral request can be submitted for your cohort
Proctored test	A supervised examination conducted under controlled conditions
Hard deadline	The final submission date beyond which submissions receive zero marks
Relative marking	Grading adjusted relative to cohort performance, per IIT Kharagpur's framework

Appendix B: Pre-Programme Checklist

- Fee payment completed
- Offer acceptance confirmed
- LMS access verified (login and test)
- Telegram app installed; joined the cohort notification channel
- Programme Coordinator's Telegram ID noted (from welcome email)
- Weekend schedule blocked for live sessions (3 hours/week)

Appendix C: Technology and Environment Requirements

Requirement	Minimum
Operating system	Windows 10+, macOS 11+, or Ubuntu 20.04+
RAM	8 GB minimum; 16 GB recommended for local model work
Internet	Stable broadband (minimum 5 Mbps)
Browser	Latest Chrome, Firefox, or Edge
Python	Python 3.9+ with pip and Jupyter Notebook/Lab
Git	Basic proficiency for version control and submissions
Webcam and microphone	Required for proctored tests

- Any additional tools or computing resources for specific labs will be communicated per module.

Appendix D: LMS and Content Access

Access Type	Duration
LMS and course content	Available for the duration of the programme. The content access end date for your cohort will be communicated via the LMS no later than 4 weeks before programme completion
Live session recordings	Available on the LMS for the duration of the programme access
Lab environments	Access details and duration communicated per module
Post-programme	Content access after programme completion, if any, is at the discretion of the programme administration and will be communicated prior to programme end

- If you have questions about access timelines, contact studentsupport@upgrad.com.

This handbook was last updated: 2026 Edition. The online version on the LMS is the most current version. Nothing in this handbook supersedes IIT Kharagpur's academic regulations (for academic matters) or upGrad's Terms and Conditions (for operational and commercial matters).